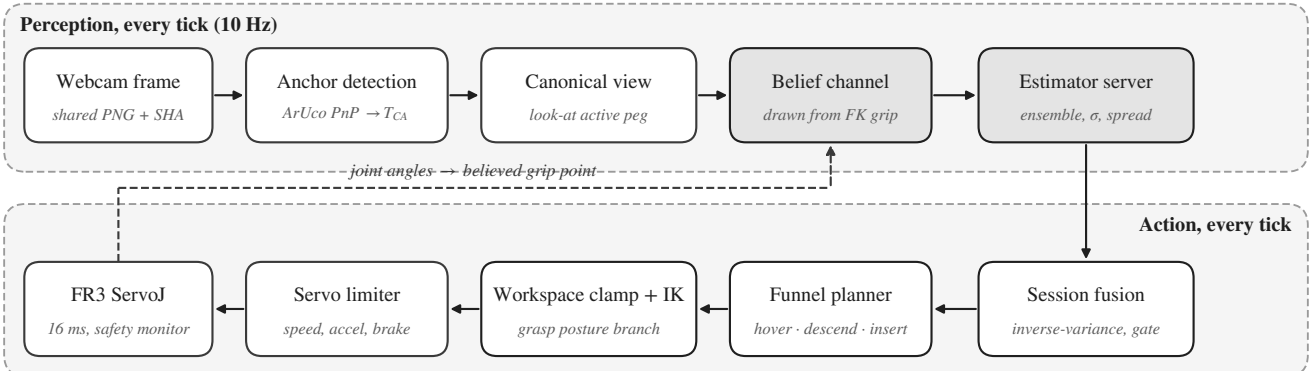
Perception, every tick (10 Hz)
Webcam frame
shared PNG + SHA
Anchor detection
ArUco PnP → T_{CA}
Canonical view
look-at active peg
Belief channel
drawn from FK grip
Estimator server
ensemble, σ, spread
joint angles → believed grip point
Action, every tick
FR3 ServoJ
16 ms, safety monitor
Servo limiter
speed, accel, brake
Workspace clamp + IK
grasp posture branch
Funnel planner
hover · descend · insert
Session fusion
inverse-variance, gate